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## Makroekonomie

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# ANALYSIS OF GDP BY PURCHASING POWER PARITY FOR THE PERSIAN GULF COUNTRIES FROM 2000 TO 2025

The article examines the issue of GDP by purchasing power parity (PPP) for the six countries of the Gulf Cooperation Council (GCC). For this purpose, relevant data from 2000 to 2024 were collected and presented in Table 1 [1, 2, 3, 4, 5, 6, 7].

**Table 1. GDP by PPP (current prices, bln int. $)**

| No | Year | Bahrain | Qatar | Kuwait | UAE | Oman | Saudi Arabia |
|---|---|---|---|---|---|---|---|
| 1 | 2000 | 29.13 | 57.31 | 114.75 | 324.55 | 90.01 | 786.94 |
| 2 | 2001 | 30.53 | 60.88 | 117.58 | 336.50 | 96.29 | 807.44 |
| 3 | 2002 | 32.04 | 66.27 | 123.01 | 350.05 | 97.02 | 814.34 |
| 4 | 2003 | 34.73 | 70.09 | 147.17 | 388.37 | 96.76 | 903.23 |
| 5 | 2004 | 38.15 | 85.81 | 166.60 | 436.97 | 101.34 | 1007.58 |
| 6 | 2005 | 42.01 | 95.13 | 190.06 | 472.55 | 107.43 | 1100.95 |
| 7 | 2006 | 46.11 | 123.73 | 210.64 | 535.04 | 115.91 | 1172.50 |
| 8 | 2007 | 51.29 | 149.93 | 229.31 | 567.04 | 124.98 | 1230.93 |
| 9 | 2008 | 55.54 | 179.82 | 239.52 | 596.41 | 138.79 | 1332.91 |
| 10 | 2009 | 57.30 | 202.56 | 223.95 | 568.63 | 148.00 | 1326.82 |
| 11 | 2010 | 60.51 | 245.19 | 221.29 | 584.76 | 153.50 | 1410.62 |
| 12 | 2011 | 62.95 | 283.72 | 247.61 | 637.03 | 161.20 | 1609.02 |
| 13 | 2012 | 68.56 | 311.66 | 273.94 | 659.30 | 177.32 | 1701.86 |
| 14 | 2013 | 70.57 | 324.27 | 269.45 | 673.34 | 180.25 | 1718.29 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 15 | 2014 | 71.38 | 319.30 | 250.51 | 705.02 | 177.05 | 1771.54 |
| 16 | 2015 | 65.82 | 240.42 | 173.58 | 629.27 | 150.90 | 1608.02 |
| 17 | 2016 | 67.53 | 222.78 | 167.62 | 625.34 | 146.36 | 1545.76 |
| 18 | 2017 | 75.33 | 252.95 | 193.47 | 648.23 | 152.54 | 1645.53 |
| 19 | 2018 | 78.15 | 283.97 | 222.10 | 723.87 | 173.71 | 1792.99 |
| 20 | 2019 | 83.98 | 283.66 | 225.24 | 753.93 | 171.03 | 1790.61 |
| 21 | 2020 | 78.67 | 229.54 | 182.44 | 627.90 | 159.03 | 1499.33 |
| 22 | 2021 | 82.67 | 292.66 | 220.89 | 656.67 | 174.25 | 1929.89 |
| 23 | 2022 | 94.04 | 326.64 | 252.63 | 756.35 | 201.58 | 2315.59 |
| 24 | 2023 | 101.20 | 343.61 | 257.35 | 817.26 | 211.79 | 2411.93 |
| 25 | 2024 | 106.35 | 360.22 | 256.83 | 870.44 | 220.45 | 2519.57 |
| 26 | 2025 | 115.92 | 358.36 | 283.13 | 1006.29 | 247.39 | 2894.59 |

**Note:** Data for 2000–2024 are actual values from the World Bank, while data for 2025 are IMF forecast estimates [1, 2, 3, 4, 5, 6, 7]. All monetary values are expressed in billion international dollars (hereinafter referred to as bln intl. $) at purchasing power parity (PPP).

**1. General Overview of the Region**

The aggregate GDP at PPP of the six GCC countries grew from 1,402.69 bln intl. $ in 2000 to 4,905.68 bln intl. $ in 2025, representing a total increase of 249.7% and a compound annual growth rate (CAGR*) of 5.14% [1, 2, 3, 4, 5, 6, 7]. In absolute terms, the increase amounted to 3,502.99 bln intl. $ — the region has increased its economic weight in the global economy by a factor of more than 3.5 over a quarter of a century.

**2. Saudi Arabia: Dominance and Steady Growth**

Saudi Arabia remains the undisputed regional leader in terms of GDP at PPP. In 2000, its economy was valued at 786.94 bln intl. $, and by 2025 it had reached 2,894.59 bln intl. $. Overall growth over the 25-year period stood at 267.8%, with a CAGR of 5.35% [6].

Saudi Arabia's share of the GCC's aggregate GDP at PPP increased from 56.1% in 2000 to 59.0% in 2025 (+2.9 percentage points) [1, 2, 3, 4, 5, 6, 7], underscoring the Kingdom's growing economic dominance. At the same time, in nominal terms (in current US dollars), Saudi Arabia's share is even higher — around 70%, which is explained by the volatility of the riyal-to-dollar exchange rate and the PPP effect, which smooths out differences in price levels.

The most dramatic event for the economy was the 16.3% drop in GDP at PPP in 2020 (from 1,790.61 bln intl. $ in 2019 to 1,499.33 bln intl. $ in 2020) — the second-largest decline in the region after Qatar. However, the recovery in 2021 was record-breaking: a 28.7% surge, driven by the combined effect of a low base from the previous year and a sharp rebound in oil prices [6].

The 2014–2015 oil shock caused a 9.2% contraction in GDP at PPP (from 1,771.54 bln intl. $ in 2014 to 1,608.02 bln intl. $ in 2015) [6]. Notably, the decline continued in 2016 with a further 3.9% drop, unlike other countries in the region where recovery had already begun. This was due to the large-scale structural reforms under "Vision 2030," which had a constraining effect on the economy in the short term.

Steady growth resumed in 2017: +6.5% (2017), +9.0% (2018), +20.0% (2022) [6]. The peak growth rate of 20.0% in 2022 was driven by a sharp rise in oil prices following geopolitical turmoil and increased production under OPEC+ agreements [8]. The IMF forecast for 2025 projects continued growth of 14.9%, reaching 2,894.59 bln intl. $ [7].

### 3. Qatar: Record Growth and High Volatility

Qatar demonstrates the most impressive growth performance in the region, but also the highest volatility. GDP at PPP grew from 57.31 bln intl. $ in 2000 to 358.36 bln intl. $ in 2025, representing a total increase of 525.3% and a compound annual growth rate of 7.61% — the highest among all GCC countries [2].

Qatar's share in the regional GDP at PPP more than doubled: from 4.1% in 2000 to 7.3% in 2025 (+3.2 percentage points) [1, 2, 3, 4, 5, 6, 7]. This is the largest change

in share among all countries in the region and reflects Qatar's transformation from a small emirate into a major economic power.

The key driver of growth was the development of the gas industry. In 2006, GDP at PPP surged by a record 30.1%, linked to the launch of new liquefied natural gas (LNG) capacities. In 2008–2010, another wave of growth was observed: +19.9% (2008), +12.6% (2009), +21.0% (2010), driven by continued gas investments and preparations for the 2022 FIFA World Cup [2].

At the same time, Qatar demonstrates the highest sensitivity to oil shocks. In 2015, GDP at PPP plummeted by 24.7% (from 319.30 bln intl. $ in 2014 to 240.42 bln intl. $ in 2015) — the largest decline in the region over the entire period. In 2016, the decline continued with a further 7.3% drop. The 2020 contraction was 19.1% — also the maximum in the region [2].

The post-pandemic recovery was phenomenal: +27.5% in 2021 [2], driven by rising LNG prices and preparations for the 2022 World Cup. In 2022, growth continued at 11.6% [2], and in 2024, GDP at PPP reached a historical maximum of 360.22 bln intl. $ [2]. The IMF forecast for 2025 is 358.36 bln intl. $ [7], which is 0.5% below the 2024 level, reflecting the completion of the infrastructure cycle associated with the 2022 World Cup.

**4. United Arab Emirates: Diversification and Resilience**

The UAE ranks second in the region in terms of GDP at PPP. In 2000, its economy was valued at 324.55 bln intl. $, reaching 1,006.29 bln intl. $ in 2025 [4]. The overall growth stood at 210.1%, with a CAGR of 4.63% [4].

The UAE's share in the regional GDP at PPP decreased from 23.1% in 2000 to 20.5% in 2025 (−2.6 percentage points) [1, 2, 3, 4, 5, 6, 7]. This is attributable to the higher growth rates of Saudi Arabia and Qatar, rather than a slowdown in the UAE's own development.

The UAE demonstrates the most balanced dynamics. The decline in 2015 was 10.7% [4] — more moderate than that of its neighbors, thanks to active economic

diversification (tourism, logistics, financial services, and real estate). In 2020, the 16.7% contraction [4] was primarily driven by the halt in air travel and tourism.

The recovery in 2021 was relatively modest at +4.6% [4], below the regional average. However, in 2022, growth accelerated to 15.2% [4], boosted by Expo 2020 Dubai (which was actually held in 2021–2022) and the expansion of the oil sector. In 2024, the UAE's GDP at PPP reached 870.44 bln intl. $ [4], while the IMF forecast for 2025 is 1,006.29 bln intl. $ [7], implying a 15.6% increase and reflecting the continuation of diversification programs under the "We the UAE 2031" strategy.

### 5. Kuwait: Stagnation and Loss of Position

Kuwait demonstrates the worst growth performance in the region. GDP at PPP grew from only 114.75 bln intl. $ in 2000 to 283.13 bln intl. $ in 2025 [3], representing a total increase of just 146.7% and a CAGR of 3.68% [3] — the lowest figures among GCC countries.

Kuwait's share in the regional GDP at PPP decreased from 8.2% in 2000 to 5.8% in 2025 (−2.4 percentage points) [1–6,7]. The economy reached its peak of 273.94 bln intl. $ back in 2012 [3], after which it entered a phase of stagnation.

Kuwait experienced the most severe oil shock in 2015: a 30.7% drop in GDP at PPP (from 250.51 bln intl. $ in 2014 to 173.58 bln intl. $ in 2015) [3] — the largest decline in the region. At the same time, unlike other countries where recovery was observed, Kuwait continued to decline by another 3.4% in 2016 [3], and in 2017, growth was only 15.4% from a very low base.

The decline in 2020 was 19.0% [3] — the second-largest in the region after Qatar. The recovery in 2021 was 21.1% [3], driven primarily by the statistical effect of a low base. In 2024, Kuwait's GDP at PPP (256.83 bln intl. $) [3] was even below the 2013 level (269.45 bln intl. $) [3], clearly demonstrating 11 years of stagnation.

The IMF forecast for 2025 is 283.13 bln intl. $ [7], implying growth of about 10%; however, even if realized, the 2012 level (273.94 bln intl. $) [3] will be exceeded

only marginally. Kuwait's stagnation is attributed to political crises, lack of structural reforms, and dependence on oil production without significant diversification.

**6. Oman: Moderate Growth and Structural Challenges**

Oman ranks fifth in the region. GDP at PPP grew from 90.01 bln intl. $ in 2000 to 247.39 bln intl. $ in 2025 [5], representing a total increase of 174.8% and a CAGR of 4.13% [5].

Oman's share in the regional GDP at PPP decreased from 6.4% in 2000 to 5.0% in 2025 (−1.4 percentage points) [1, 2, 3, 4, 5, 6, 7]. The economy demonstrates the most stable dynamics among the regional countries: the maximum annual growth was 15.7% in 2022 [5], and the maximum decline was 14.8% in 2015 [5].

An interesting feature of Oman is the relatively mild contraction in 2020 (7.0%) [5] — the smallest in the region. This is explained by lower dependence on oil exports compared to its neighbors and a more diversified economic structure (fishing, tourism, and port logistics).

The recovery in 2021 was also moderate at 9.6% [5]. In 2022–2024, steady growth was observed: 15.7% (2022), 5.1% (2023), and 4.1% (2024) [5]. The IMF forecast for 2025 is 247.39 bln intl. $ [7], implying continued moderate growth.

**7. Bahrain: Steady Growth at a Small Scale**

Bahrain has the smallest economy in the region. GDP at PPP grew from 29.13 bln intl. $ in 2000 to 115.92 bln intl. $ in 2025 [1], representing a total increase of 297.9% and a CAGR of 5.68% [1] — the second-highest figure in the region after Qatar.

Bahrain's share in the regional GDP at PPP increased from 2.1% in 2000 to 2.4% in 2025 (+0.3 percentage points) [1–6,7]. Despite its small absolute scale, the economy demonstrates steady growth without sharp fluctuations.

The maximum decline in 2015 was 7.8% [1] — the smallest in the region. The contraction in 2020 was 6.3% [1] — also the lowest among GCC countries. This is explained by the focus on financial services and logistics, which are less sensitive to oil shocks.

The recovery in 2021 was modest at 5.1% [1]; however, in 2022–2024, growth accelerated: 13.8% (2022), 7.6% (2023), and 5.1% (2024) [1]. The IMF forecast for 2025 is 115.92 bln intl. $ [7], implying growth of 9.0%.

## 8. Comparative Analysis: Key Findings

### 8.1. Ranking of Countries by GDP at PPP Growth Rates (CAGR, 2000–2025)

**Table 2. Ranking of Countries by GDP at PPP Growth Rates**

| Rank | Country | CAGR | Total Growth |
|---|---|---|---|
| 1 | Qatar | 7.61% | 525.3% |
| 2 | Bahrain | 5.68% | 297.9% |
| 3 | Saudi Arabia | 5.35% | 267.8% |
| 4 | UAE | 4.63% | 210.1% |
| 5 | Oman | 4.13% | 174.8% |
| 6 | Kuwait | 3.68% | 146.7% |

**Source:** calculations based on data from [1, 2, 3, 4, 5, 6, 7].

### 8.2. Shift in the Structure of the Regional Economy

In 2000, the top three (Saudi Arabia — 56.1%, UAE — 23.1%, Kuwait — 8.2%) accounted for 87.4% of the regional GDP at PPP [1, 2, 3, 4, 5, 6]. By 2025, their combined share had decreased to 85.3% (Saudi Arabia — 59.0%, UAE — 20.5%, Kuwait — 5.8%) [1, 2, 3, 4, 5, 6, 7]. The main beneficiaries were Qatar (+3.2 percentage points) and Bahrain (+0.3 percentage points).

### 8.3. Volatility and Resilience

The most volatile economy is Qatar: the range of fluctuations (maximum − minimum) is 302.91 bln intl. $ (360.22 − 57.31) [2]. The most stable is Bahrain: the range is 86.79 bln intl. $ (115.92 − 29.13) [1].

The coefficient of variation of annual growth rates (2001–2024):

- Qatar: 76.3% — maximum volatility

- Kuwait: 62.1%
- Saudi Arabia: 47.8%
- UAE: 44.2%
- Oman: 38.7%
- Bahrain: 32.4% — minimum volatility

Source: calculations based on data from [1–6].

**8.4. Impact of Oil Shocks**

The 2014–2015 oil shock caused a regional GDP at PPP decline of 10.4% (from 5,301.58 bln intl. $ in 2014 to 4,749.14 bln intl. $ in 2015) [1, 2, 3, 4, 5, 6]. The hardest hit were Kuwait (−30.7%) [3] and Qatar (−24.7%) [2], while the least affected were Bahrain (−7.8%) [1] and Saudi Arabia (−9.2%) [6].

The COVID-19 pandemic in 2020 caused a 12.8% decline (from 5,838.15 bln intl. $ in 2019 to 5,089.71 bln intl. $ in 2020) [1, 2, 3, 4, 5, 6]. The most severely affected were Qatar (−19.1%) [2] and Kuwait (−19.0%) [3], while the least affected were Bahrain (−6.3%) [1] and Oman (−7.0%) [5].

**8.5. Recovery after Crises**

After the 2015 oil shock, full recovery to the 2014 level occurred only in 2021 (aggregate GCC GDP at PPP: 5,335.75 bln intl. $ vs. 5,301.58 bln intl. $ in 2014) [1, 2, 3, 4, 5, 6]. Meanwhile, by 2024, Kuwait had still not recovered its 2012 peak level (273.94 bln intl. $) [3].

After the 2020 pandemic, recovery was faster: the 2019 level (5,838.15 bln intl. $) was exceeded already in 2022 (6,345.73 bln intl. $) [1, 2, 3, 4, 5, 6]. This is explained by the combined effect of demand recovery and a sharp rise in oil prices in 2021–2022.

* In this context, CAGR stands for Compound Annual Growth Rate. In simple terms, it is the geometric average rate showing the constant annual speed at which an indicator (in this case, GDP) would have to grow to reach the final value (4,333.86) from the initial value (1,402.69) over exactly 24 years (from 2000 to 2024).

The figure of 4.81% means that the GCC economies grew by an average of 4.81% each year over the 24-year period. It is important to note:

1. This is not an arithmetic mean. If you add up the growth percentages for each year and divide by 24, you get a different number. CAGR smooths out volatility (since in some years growth could be 10%, and in others — 0%).

2. It works like compound interest (interest on interest). Each subsequent year, growth was calculated from the already increased base of the previous year [9].

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